

Austin Lutterbach

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EDUCATION

Cornell University

- Degree: Masters of Engineering
- Concentration: Applied Operations Research

GPA 3.80
Aug 2020 - May 2021

Purdue University

- Degree: BS in Math and Statistics

GPA 3.73
Aug 2017 - May 2020

SKILLS

Frameworks: Python (8+ yrs), LaTeX (8+ yrs), Markdown (8+ yrs), LlamaIndex (2+ yrs)

Software/COTS: Docker (3+ yrs), Linux (3+ yrs), Git (3+ yrs), Cursor (1 yr)

RELEVANT EXPERIENCE

Booz Allen Hamilton, Machine Learning Engineer/Scientist

Jan 2024 - Present

- Fine-tuned a local LLM (gemma 2b) to be used by combat medics to transcribe critical information in chaotic environments.
- Co-authored white paper detailing current trends in LLM observability functionalities.
- Developed adversarial patch attacks on object detection models to drive misclassifications.
- Developed poison patrol, a software tool that combines multiple methods of detecting poison in training data into one cohesive detection mechanism, building on latest research practices.
- Developed agentic workflows to help streamline fraud detection for USPS.
- Developing agentic workflows for OT cybersecurity applications.

Cobec Consulting, Data Scientist

Aug 2021 - Jan 2024

- Improved failure forecast accuracy by leveraging industry-standard modeling tools (scipy, scikit-learn, tensorflow), reducing model RMSE by 35% compared to existing FAA models.
- Predicted critical parts failures in the NAS using LSTM deep learning models on time-series failure data.
- Assisted in developing Java and SQL-based database systems to parse live flight data for statistical analysis.
- Nominated for ICEAA 2023 Junior Analyst of the Year Award for analyses leading to better-informed FAA infrastructure decisions, saving millions in avoided risk mitigation costs.

Utiliz, Machine Learning Engineer (Startup)

Jan 2022 - Sep 2023

- Led the end-to-end development of a machine learning project predicting maintenance needs for tens of thousands of water pipes in Evansville.
- Streamlined data preprocessing, model training, deployment, and maintenance using AWS Glue and SageMaker along with Docker.
- Engineered a Graph Convolutional Neural Network (GCN) with PyTorch to better capture pipe network structure and enhanced failure prediction accuracy to 79%.
- Utilized an OCR parser to extract and vectorize textual data from handwritten legal documents with Llama 2 LLM to generate automated legal reports based on historical report structures.
- Implemented a dual-layer verification mechanism by utilizing another LLM to scrutinize and validate the accuracy of the generated reports against a comprehensive database of water utility documentation, ensuring the production of high-quality and error-free outputs with optional human oversight in the final review stage.

Laboratory for Computational Anthroinformatics, Purdue University, Research Assistant

Feb 2019 – Aug 2020

- Developed machine learning models and performed statistical analyses on archaeological data using R, exploring ancient human dietary and migratory patterns
- Conducted literature review on morphometrics and implemented methods such as generalized Procrustes into software package called “GUImorph” to help archaeologists streamline morphometric calculations on archaeological specimens.

PROJECTS

Summer of Open AI Research, EleutherAI

August 2026

- Supported the UniverseTBD research group in project titled “Effective Rank and Semantic Compression in Foundation Models”
- Researched methods to evaluate the “Platonic Representation Hypothesis” by trying to decouple properties of the underlying manifold with the performance of alignment evaluations.
- Discovered connection between local prediction accuracy on the embedding manifold and label curvature relative to the bending of latent space.
- Authored white paper on these findings and submitted to ML4PS workshop as first author.